

Notes: Duflo (AER, 2001)

Schooling and Labor Market Consequences of School Construction in Indonesia: Evidence from an Unusual Policy Experiment

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1 Big picture

- **Question.** Did a very large primary-school construction program in Indonesia increase schooling and later wages? If it did, what does that imply about the causal return to education in a developing country?
- **Policy experiment.** Between 1973-1974 and 1978-1979, the Indonesian government built more than 61,000 primary schools through the Sekolah Dasar INPRES program. The scale was enormous: on average roughly two schools per 1,000 children aged 5 to 14.
- **Identification idea.** Exposure varies along two dimensions. First, districts received different numbers of schools. Second, cohorts differed in age when the program began. Young children in high-construction districts were exposed; older cohorts in the same districts were not.
- **Core design.** The paper combines region-of-birth fixed effects, year-of-birth fixed effects, and interactions between cohort exposure and regional program intensity. This is a cohort-by-region difference-in-differences design.
- **Main schooling effect.** Each new school per 1,000 children increased schooling by roughly 0.12 to 0.19 years for the fully exposed cohorts.
- **Main wage effect.** Each new school per 1,000 children increased later wages by about 1.5 to 2.7 percent for the exposed cohorts.
- **Return to schooling.** Dividing the reduced-form wage effect by the first-stage schooling effect gives instrumental-variable returns to education of about 6.8 to 10.6 percent.
- **Why this matters.** The paper provides early and influential causal evidence that expanding school infrastructure can raise education and labor-market outcomes in a developing country.
- **Scope of the effect.** The program mainly increased completion of primary school and some lower secondary schooling, rather than higher education.
- **Cost-benefit result.** Under reasonable growth assumptions, the social returns to the program are positive and large; the internal rate of return can exceed conventional public investment benchmarks.
- **Economic takeaway.** School supply constraints matter. Building schools can increase human capital and raise wages, and an appropriately targeted large-scale school construction program can pass a social cost-benefit test.

2 Incremental contribution of this paper

- **From correlations to quasi-experimental evidence.** Before this paper, many estimates of returns to schooling in developing countries were based on cross-sectional correlations between education and wages. Duflo uses a policy shock to build a much stronger causal design.
- **A large-scale school-supply shock.** The paper studies one of the largest school-construction programs on record. This shifts the literature from small interventions or observational comparisons to a nationwide infrastructure expansion.
- **Two-dimensional exposure design.** The central contribution is the interaction of *where* a person was born with *how old* the person was when the program began. Regions with more schools identify spatial exposure; cohorts identify whether individuals were young enough to benefit.
- **A natural instrument for education.** The interactions between program intensity and cohort exposure create instruments for years of schooling. This allows the paper to estimate the causal wage return to schooling, not merely the reduced-form effect of school construction.
- **Testable validity checks.** The design implies that cohorts too old to benefit should show no program effect. The paper uses these older cohorts as a control experiment and visually displays the cohort pattern in Figure 1 and Figure 3.

- **Distinguishing quantity from quality.** Because school construction could in principle affect school quality, the paper checks pupil-teacher ratios, education-level effects, and subgroup estimates. This makes the IV interpretation more credible.
- **Cost-benefit integration.** The paper does not stop at earnings effects. It uses construction, teacher, and operating-cost data to ask whether the program’s social benefits exceeded costs.
- **Methodological legacy.** The paper became a template for using large public programs as quasi-experiments in development economics, especially when exposure can be defined by age/cohort and geography.

3 Data and measurement

3.1 Individual-level data

- The main individual data source is the 1995 intercensal survey of Indonesia, SUPAS.
- The sample focuses on men born between 1950 and 1972.
- The full sample has 152,989 individuals; the wage-earner sample has about 60,633 individuals.
- The main individual outcomes are completed years of education and log wages or earnings.
- Mean completed schooling in the full sample is 7.98 years; mean schooling in the valid wage sample is 9.00 years.

Interpretation: The sample is old enough in 1995 to observe completed schooling and labor-market outcomes, while still young enough that some cohorts were affected by the 1970s construction program.

3.2 Program intensity

- Program intensity is the number of INPRES schools constructed in a district, scaled by the number of children aged 5 to 14.
- The average program intensity is about 1.98 schools per 1,000 children in the individual sample.
- District-level data indicate an average of 222 INPRES schools constructed per district and 2.34 schools per 1,000 children.
- High-program regions have more schools built per child than low-program regions, but high-program regions initially had lower enrollment.

Interpretation: The program was not randomly assigned. It was intentionally targeted toward low-enrollment areas. That is why the design uses region fixed effects, cohort fixed effects, and pre-program controls.

3.3 Cohort exposure

- Children aged 2 to 6 in 1974 were young enough to be fully exposed to the new primary schools.
- Children aged 12 to 17 in 1974 were essentially too old to benefit because they should already have completed primary school.
- Some intermediate cohorts may be partially exposed.
- Cohort exposure is measured by age in 1974 and interacted with program intensity in the region of birth.

Interpretation: The cohort dimension is essential. It creates within-region comparison groups, so the paper does not simply compare poorer high-program regions to richer low-program regions.

3.4 Region of birth

- The paper links individuals to program intensity using district or region of birth.
- Region of birth is used instead of region of education because school attendance location could respond to the program.
- The paper notes that region of birth and region of education are highly correlated: most children in the relevant survey evidence were still living in their birth district at age 12.

Interpretation: Region of birth is predetermined relative to school construction because all individuals were born before the program began. This limits endogenous migration concerns.

3.5 Wage and earnings outcomes

- The main wage outcome is log hourly wage for wage earners.
- The paper also studies log monthly earnings.
- Because only wage earners report wages, the paper examines sample selection by estimating wage-sector participation and by imputing earnings for self-employed workers from SUSENAS occupation cells.

Interpretation: Sample selection matters because schooling can change the probability of working for a wage. The paper therefore checks whether the returns are driven only by selection into the wage-earning sample.

3.6 Cost-benefit inputs

- The paper uses information on construction costs, teacher salaries, teacher training, recurrent costs, school lifetime, and deadweight costs of public funds.
- It compares annual and discounted benefits to the cost of building, staffing, and maintaining the schools.
- The cost-benefit exercise uses wage effects as a proxy for productivity gains.

Interpretation: The cost-benefit exercise is intentionally cautious. Wage gains do not include other potential benefits of education, such as fertility, child health, civic participation, or intergenerational effects.

4 Models and empirical specifications

4.1 Simple difference-in-differences idea

Let S_{ijk} be education of individual i born in region j and year k . Let P_j be program intensity and T_i indicate the treated young cohort.

$$S_{ijk} = c + \alpha_j + \beta_k + \gamma(P_j T_i) + \delta(C_j T_i) + \varepsilon_{ijk}. \quad (1)$$

- α_j are region-of-birth fixed effects.
- β_k are year-of-birth fixed effects.

- $P_j T_i$ is the interaction of regional program intensity and cohort exposure.
- $C_j T_i$ allows baseline region characteristics to affect younger cohorts differently.

Interpretation: The coefficient γ compares younger and older cohorts across high- and low-program regions. It is the effect of one additional school per 1,000 children on completed schooling.

4.2 Unrestricted cohort-by-cohort specification

The paper estimates cohort-specific effects using interactions between program intensity and age-in-1974 dummies:

$$S_{ijk} = c + \alpha_j + \beta_k + \sum_{l=2}^{23} (P_j d_{il}) \gamma_l + \sum_{l=2}^{23} (C_j d_{il}) \delta_l + \varepsilon_{ijk}. \quad (2)$$

Interpretation: Each γ_l estimates the program effect for people who were age l in 1974. This specification is the basis for Figure 1 and Figure 3. It is a visual and statistical placebo test: older cohorts should have coefficients near zero.

4.3 Restricted cohort specification

For efficiency, the paper imposes the restriction that cohorts too old to benefit have zero effect and focuses on cohorts aged 2 to 12:

$$S_{ijk} = c + \alpha_j + \beta_k + \sum_{l=2}^{12} (P_j d_{il}) \gamma_l + \sum_{l=2}^{12} (C_j d_{il}) \delta_l + \varepsilon_{ijk}. \quad (3)$$

Interpretation: This uses the theoretical timing of primary school exposure. Children age 13 or older in 1974 should not be directly affected because they should have left primary school before new schools opened.

4.4 CDF / level-of-education specification

To identify the level of education at which the program mattered, the paper estimates a linear probability model for completing at least m years of education:

$$S_{ijkm} = c + \alpha_j + \beta_k + (P_j T_i) \kappa_m + \varepsilon_{ijkm}. \quad (4)$$

Interpretation: The coefficients κ_m show where in the schooling distribution the program had bite. If the program mainly increased primary completion, effects should rise up to around six years of schooling and then flatten or decline.

4.5 Reduced-form wage specification

The same design is used with log wages:

$$Y_{ijk} = c + \alpha_j + \beta_k + \gamma(P_j T_i) + \delta(C_j T_i) + \varepsilon_{ijk}, \quad (5)$$

where Y_{ijk} is log hourly wage or log monthly earnings.

Interpretation: This is the reduced-form effect of school construction on later earnings. It captures all channels through which the program affects wages, including schooling and any other program-induced changes.

4.6 Instrumental variables return to education

The wage equation for returns to education is

$$Y_{ijk} = c + \rho S_{ijk} + \alpha_j + \beta_k + \delta(C_j T_i) + u_{ijk}, \quad (6)$$

where S_{ijk} is instrumented with program exposure interactions.

Interpretation: The IV estimate ρ is the causal return to education for the compliers induced to obtain more schooling by the INPRES program. It is a local average treatment effect, mostly for primary schooling.

4.7 Cost-benefit calculation

The annual cost is written as

$$C(t) = rK + rTC + 1.25W(t), \quad (7)$$

where K is construction cost, TC is teacher training cost, $W(t)$ is teacher salary cost, and r is the discount rate.

Interpretation: The formula separates capital cost, training cost, and recurrent teacher costs. The factor 1.25 accounts for recurrent non-salary costs. Benefits are projected from wage gains attributable to the program.

5 Identification and empirical design

5.1 Core identifying variation

- The program was rolled out nationally but with different intensity across regions.
- The intensity was higher in regions with lower pre-program enrollment.
- Cohorts young enough to attend primary school after construction are treated; older cohorts in the same regions are not.
- The identifying comparison is therefore a difference in cohort trends between high- and low-program regions.

Interpretation: The design is not a randomized controlled trial, but the cohort-by-region structure is powerful because it uses older cohorts to absorb fixed differences across regions.

5.2 Main identifying assumption

The key assumption is that, absent the INPRES program, the change in education and wages between older and younger cohorts would not have differed systematically between high- and low-program regions after controls.

Interpretation: This is a parallel-trends assumption across regions and cohorts. The paper tests its implications using older cohorts and by controlling for initial enrollment and other programs.

5.3 Why the targeting rule matters

- Schools were allocated partly based on the number of children not enrolled in primary school.
- This means high-program regions were initially more disadvantaged.
- A simple cross-sectional comparison would be biased.
- The regression in Table 2 verifies that the allocation rule strongly predicts the number of schools.

Interpretation: The policy targeting is not a flaw if handled correctly. It provides variation in program intensity but requires fixed effects and controls to avoid confusing program effects with mean reversion.

5.4 Placebo and cohort-pattern checks

- Children aged 13 or older in 1974 should not benefit directly.
- Figure 1 shows coefficients near zero for older cohorts and positive effects for younger cohorts.
- Figure 3 shows a similar pattern for education and wages.
- Table 4 includes a control experiment using older cohorts.

Interpretation: The cohort pattern is a central validation exercise. The timing of effects follows the timing of exposure to primary school construction.

5.5 Quality versus quantity concern

- The IV return to education assumes that the program affects wages primarily through years of education, not through changes in school quality.
- The paper checks pupil-teacher ratios and finds no evidence that quality changed systematically with program intensity.
- Effects are concentrated at primary education levels, which matches the targeted infrastructure.

Interpretation: If the program reduced or improved quality, the IV estimate would combine quantity and quality effects. The paper argues the evidence favors a quantity interpretation.

5.6 Sample-selection concern

- Wages are observed only for wage earners.
- Education can change wage-sector participation.
- Table 7 estimates effects on wage-sector participation and uses imputed earnings for the self-employed.

Interpretation: The wage-earner estimates are not the only evidence. The completed-sample exercise suggests that returns may be somewhat lower outside the wage sector, but the causal schooling effect remains economically meaningful.

5.7 General equilibrium concern

- A program that raises the national supply of primary graduates can lower returns to education nationwide.
- Region fixed effects and cohort fixed effects cannot recover this common general equilibrium component.
- The estimated wage effect is therefore not a complete national welfare effect if returns changed for everyone.

Interpretation: This is one of the paper’s important limitations. The paper estimates relative exposure effects across regions, not the full national equilibrium effect of changing the supply of educated labor.

6 Section-by-section study guide

- **Introduction.** Motivates the problem of estimating causal returns to education and introduces the INPRES program as a policy experiment.
- **Program and data.** Describes SUPAS, district-level program intensity, school allocation, and the age-by-region exposure design.
- **Effect on education.** Estimates how schooling changes for exposed cohorts in high-program regions and checks the cohort pattern.
- **Effect on wages.** Applies the same reduced-form design to wages.
- **Returns to education.** Uses program exposure as an instrument for schooling in wage equations.
- **Cost-benefit analysis.** Converts the reduced-form wage gains into social benefits and compares them with program costs.
- **Conclusion.** Interprets the returns to education and discusses generalizability.

7 Tables and figures

7.1 Table 1: Descriptive Statistics + Table 2: The Allocation of Schools

Read:

- Table 1 reports individual, district, and auxiliary IFLS summary statistics.
- The full SUPAS sample has 152,989 men born between 1950 and 1972; the wage sample has about 60,663 men.
- Average schooling is 7.98 years in the full sample and 9.00 years in the wage sample.
- Average program intensity is 1.98 INPRES schools per 1,000 children in the individual sample.
- High-program regions received about 2.44 schools per 1,000 children; low-program regions received about 1.54.
- Table 2 shows that allocation is strongly related to the number of children and to the pre-program non-enrollment rate.
- The R^2 in Table 2 is high, consistent with a policy rule allocating schools to underserved regions.

Detailed interpretation: Table 1 establishes why simple cross-sectional comparisons would be misleading. High-program areas have lower initial school access, so high program intensity is associated with underlying disadvantage. Table 2 formalizes this: the allocation of schools is not random but follows observable need-based variables. The empirical strategy must therefore rely on cohort exposure within regions and controls for pre-existing conditions.

Economic message: The program was redistributive and targeted. That makes it substantively important, but it also means causal inference requires careful difference-in-differences logic rather than naive comparisons.

TABLE 1—DESCRIPTIVE STATISTICS

	Mean
<i>Panel A: Individual Level Means</i>	
Education (whole sample $N = 152,989$)	7.98
Education (sample with valid wage data $N = 60,663$)	9.00
INPRES schools built per 1,000 children	1.98
INPRES schools built per 1,000 children (sample with valid wage data)	1.89
INPRES schools built per 1,000 children (High program regions)	2.44
INPRES schools built per 1,000 children (Low program regions)	1.54
Log(hourly wage)	6.87
Monthly earnings (SUPAS 1995), thousands Rupiah	13
Monthly earnings (SUSENAS 1993) of wage earners, thousands Rupiah	205
Monthly earnings (SUSENAS 1993) of self-employed individuals, thousands Rupiah	152
<i>Panel B: District Level Means ($N = 293$)</i>	
INPRES schools constructed (1973–1974 to 1978–1979)	222
INPRES schools constructed per 1000 children (1973–1974 to 1978–1979)	2.34
Number of teachers in 1973–1974	1,530
Number of teachers in 1978–1979	2,082
Number of schools in 1973–1974	219
Fraction of the population attending school in 1971 (Census)	0.174
Enrollment rate in primary school in 1973 (Ministry of Education and Culture)	0.68
<i>Panel C: Indonesian Family Life Survey, Individuals Born Between 1950 and 1972 (all numbers are in percentages)</i>	
Proportion of individuals having migrated between birth and age 12	8.5
Proportion of people having repeated at least one grade in primary school	20.0
Proportion of people completing more than primary having repeated at least one grade in primary school	6.0
Proportion of individuals having attended primary school after age 12 (estimated)	15.8
Proportion of individuals having attended primary school after age 13 (estimated)	6.8
Proportion of individuals born 1950–1961, completing primary or less, who left school after 1974	2.8
Proportion of individuals born 1962–1966, completing primary or less, who left school after 1974	24.5

(a) Table 1: Descriptive Statistics

TABLE 2 THE ALLOCATION OF SCHOOLS

	Log(INPRES schools) ^a
Log of number of children aged 5–14 in the region	0.78 (0.027)
Log(1 – enrollment rate in primary school in 1973) ^b	0.12 (0.038)
Number of observations	255
R^2	0.78

Notes: Standard errors are in parentheses.

^a The dependent variable is the log of the number of INPRES schools built between 1973 and 1978.

^b The enrollment rate in primary school is the number of children enrolled in primary school in 1973 (obtained from the Ministry of Education and Culture) divided by the number of children aged 5–14 in the region in 1973.

(b) Table 2: The Allocation of Schools

7.2 Table 3: Means of Education and Log(Wage) by Cohort and Level of Program Cells

Read:

- Table 3 is the simple two-by-two difference-in-differences illustration.
- In high-program regions, young exposed cohorts have 8.49 years of schooling versus 8.02 years for older cohorts, a gain of 0.47 years.
- In low-program regions, young cohorts have 9.76 years versus 9.40 years, a gain of 0.36 years.
- The difference-in-differences is 0.12 years of schooling.
- For wages, the corresponding difference-in-differences is 0.026 in logs.

- The control experiment with older cohorts produces smaller education effects.

Detailed interpretation: The table shows the empirical design in its simplest form. High-program regions were initially less educated and lower wage, so the level comparison is not causal. What matters is the difference in changes across cohorts. The young-vs-old cohort change is larger in high-program regions, exactly what the policy timing predicts. The table also shows that the raw Wald estimate of returns to education comes from the ratio of the wage effect to the schooling effect.

Economic message: This table is the intuition for the entire paper: school construction is identified by a cross-region difference in cohort growth, not by cross-region levels.

	Years of education			Log(wages)		
	Level of program in region of birth			Level of program in region of birth		
	High (1)	Low (2)	Difference (3)	High (4)	Low (5)	Difference (6)
<i>Panel A: Experiment of Interest</i>						
Aged 2 to 6 in 1974	8.49 (0.043)	9.76 (0.037)	-1.27 (0.057)	6.61 (0.0078)	6.73 (0.0064)	-0.12 (0.010)
Aged 12 to 17 in 1974	8.02 (0.053)	9.40 (0.042)	-1.39 (0.067)	6.87 (0.0085)	7.02 (0.0069)	-0.15 (0.011)
Difference	0.47 (0.070)	0.36 (0.038)	0.12 (0.089)	-0.26 (0.011)	-0.29 (0.0096)	0.026 (0.015)
<i>Panel B: Control Experiment</i>						
Aged 12 to 17 in 1974	8.02 (0.053)	9.40 (0.042)	-1.39 (0.067)	6.87 (0.0085)	7.02 (0.0069)	-0.15 (0.011)
Aged 18 to 24 in 1974	7.70 (0.059)	9.12 (0.044)	-1.42 (0.072)	6.92 (0.0097)	7.08 (0.0076)	-0.16 (0.012)
Difference	0.32 (0.080)	0.28 (0.061)	0.034 (0.098)	0.056 (0.013)	0.063 (0.010)	0.0070 (0.016)

Notes: The sample is made of the individuals who earn a wage. Standard errors are in parentheses.

Table 3: Means of Education and Log(Wage) by Cohort and Level of Program Cells

7.3 Table 4: Effect of the Program on Education and Wages + Figure 1: Coefficients of Age-in-1974 Interactions in the Education Equation

Read:

- Table 4 estimates the main reduced-form effect using a treated cohort indicator for individuals aged 2 to 6 in 1974.
- The education effect ranges from about 0.124 to 0.188 years in the whole sample.
- In the wage-earner sample, the education effect ranges from about 0.196 to 0.259 years.
- The log hourly wage effect ranges from 0.0147 to 0.0270.
- The older-cohort control experiment has much smaller coefficients, consistent with the timing of exposure.
- Figure 1 plots cohort-specific education effects. The effects are near zero for older cohorts and rise for younger cohorts, especially after age 10 or below.

Detailed interpretation: Table 4 is the first main regression evidence. Adding controls for pre-program enrollment and the water/sanitation program generally increases the estimated effect rather than eliminating it. This suggests the schooling result is not driven by omitted development programs moving in the same direction. Figure 1 is important because it shows that the effect appears exactly where the policy should matter: among cohorts young enough to use the new schools.

Economic message: School construction changed educational attainment for the cohorts who could actually use the schools. The timing pattern is the strongest simple validation of the identification strategy.

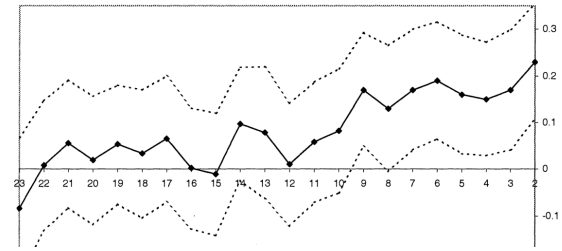
TABLE 4—EFFECT OF THE PROGRAM ON EDUCATION AND WAGES: COEFFICIENTS OF THE INTERACTIONS BETWEEN COHORT DUMMIES AND THE NUMBER OF SCHOOLS CONSTRUCTED PER 1,000 CHILDREN IN THE REGION OF BIRTH

	Observations	Dependent variable					
		Years of education			Log(hourly wage)		
		(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Experiment of Interest: Individuals Aged 2 to 6 or 12 to 17 in 1974</i>							
<i>(Youngest cohort: Individuals ages 2 to 6 in 1974)</i>							
Whole sample	78,470	0.124 (0.0250)	0.15 (0.0260)	0.188 (0.0289)			
Sample of wage earners	31,061	0.196 (0.0424)	0.199 (0.0429)	0.259 (0.0499)	0.0147 (0.00729)	0.0172 (0.00737)	0.0270 (0.00850)
<i>Panel B: Control Experiment: Individuals Aged 12 to 24 in 1974</i>							
<i>(Youngest cohort: Individuals ages 12 to 17 in 1974)</i>							
Whole sample	78,488	0.0093 (0.0260)	0.0176 (0.0271)	0.0075 (0.0297)			
Sample of wage earners	30,225	0.012 (0.0474)	0.024 (0.0481)	0.079 (0.0555)	0.0031 (0.00798)	0.00399 (0.00809)	0.0144 (0.00915)
<i>Control variables:</i>							
Year of birth*enrollment rate in 1971		No	Yes	Yes	No	Yes	Yes
Year of birth*water and sanitation program		No	No	Yes	No	No	Yes

Notes: All specifications include region of birth dummies, year of birth dummies, and interactions between the year of birth dummies and the number of children in the region of birth (in 1971). The number of observations listed applies to the specification in columns (1) and (4). Standard errors are in parentheses.

(a) Table 4: Program effects on education and wages

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(b) Figure 1: Age-in-1974 interactions in education equation

7.4 Table 5: Effect of the Program on Education and Wages by Age in 1974

Read:

- Table 5 reports unrestricted age-specific interaction coefficients.
- For education, coefficients are small or near zero for older children and become positive for younger cohorts.
- The effect is especially visible for ages 2 to 9 in 1974.
- The F-statistics indicate joint significance of the cohort-by-program interactions.
- Wage effects are noisier but broadly follow the cohort exposure pattern.

Detailed interpretation: The table relaxes the simple treatment indicator and lets the data show the full age profile. This is useful because school exposure is not a sharp binary in reality; children of different ages were partly or fully exposed. The rising profile for younger ages gives credibility to the program interpretation. If unrelated regional trends were driving the result, there would be no reason for coefficients to line up with primary-school eligibility.

Economic message: The program effect is not a mechanical regression artifact. It has the life-cycle pattern implied by the institutional timing of school construction.

7.5 Table 6: Program Effect and Returns to Education by Categories of Region of Birth

Read:

- Table 6 studies heterogeneity by density, poverty, and pre-program education.
- The program has a larger education effect in low-density regions, consistent with the idea that new schools reduced travel distance and physical access constraints.
- Effects are weaker or absent in dense regions, where additional schools may not have changed access as much.
- The returns estimates are meaningful only where there is a first stage.
- The table checks whether results are driven only by particular regional types.

TABLE 5—EFFECT OF THE PROGRAM ON EDUCATION AND WAGES: COEFFICIENTS OF THE INTERACTIONS BETWEEN DUMMIES INDICATING AGE IN 1974 AND THE NUMBER OF SCHOOLS CONSTRUCTED PER 1,000 CHILDREN IN REGION OF BIRTH

Age in 1974	Dependent variable: years of education						Dependent variable: log(hourly wage)		
	Whole sample			Sample of wage earners					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
12	-0.035 (0.047)	-0.025 (0.048)	0.002 (0.054)	-0.040 (0.077)	-0.010 (0.078)	0.009 (0.091)	0.016 (0.013)	0.019 (0.013)	0.027 (0.015)
11	0.011 (0.046)	0.025 (0.047)	0.018 (0.051)	0.008 (0.073)	0.014 (0.074)	-0.003 (0.083)	-0.014 (0.012)	-0.013 (0.013)	-0.009 (0.014)
10	0.059 (0.047)	0.049 (0.049)	0.078 (0.054)	0.10 (0.075)	0.092 (0.076)	0.13 (0.090)	0.0036 (0.013)	0.0042 (0.013)	0.0059 (0.015)
9	0.14 (0.039)	0.14 (0.041)	0.15 (0.044)	0.067 (0.065)	0.063 (0.066)	0.17 (0.077)	0.0095 (0.011)	0.010 (0.011)	0.018 (0.013)
8	0.088 (0.049)	0.11 (0.050)	0.11 (0.054)	0.19 (0.078)	0.20 (0.079)	0.28 (0.089)	0.019 (0.013)	0.021 (0.013)	0.027 (0.015)
7	0.12 (0.044)	0.14 (0.046)	0.16 (0.051)	0.11 (0.072)	0.13 (0.073)	0.16 (0.084)	-0.0095 (0.012)	-0.0049 (0.012)	0.0066 (0.014)
6	0.14 (0.042)	0.17 (0.044)	0.26 (0.049)	0.23 (0.070)	0.23 (0.070)	0.32 (0.084)	0.011 (0.012)	0.013 (0.012)	0.018 (0.014)
5	0.10 (0.043)	0.13 (0.045)	0.13 (0.050)	0.14 (0.075)	0.16 (0.075)	0.27 (0.088)	0.021 (0.013)	0.023 (0.013)	0.052 (0.015)
4	0.11 (0.039)	0.12 (0.041)	0.18 (0.046)	0.19 (0.069)	0.19 (0.069)	0.29 (0.082)	0.019 (0.012)	0.020 (0.012)	0.038 (0.014)
3	0.11 (0.044)	0.14 (0.046)	0.20 (0.053)	0.15 (0.079)	0.17 (0.080)	0.30 (0.097)	0.0079 (0.013)	0.013 (0.014)	0.027 (0.016)
2	0.14 (0.041)	0.19 (0.043)	0.19 (0.049)	0.20 (0.073)	0.22 (0.074)	0.25 (0.088)	0.016 (0.012)	0.023 (0.013)	0.040 (0.015)

Notes: All specifications include region of birth dummies, year of birth dummies, and interactions between the year of birth dummies and the number of children in the region of birth (in 1971). Standard errors are in parentheses.
^a The control group is comprised of individuals aged 13–24 in 1974.

Table 5: Effect of the Program on Education and Wages by Age in 1974

Detailed interpretation: This table connects the estimates to the mechanism. If the program works by relaxing school availability and travel-cost constraints, low-density areas should benefit more. The table supports that interpretation. It also helps address quality concerns: in subsamples where the program has no education effect, it also has little wage effect, suggesting wage gains are linked to schooling quantity rather than an independent shock to wage opportunities.

Economic message: The program worked most strongly where school access was most binding. This turns the reduced-form result into a more plausible economic mechanism.

7.6 Figure 2: Difference in Differences in CDF of Education + Figure 3: Age-in-1974 Interactions in Wage and Education Equations

Read:

- Figure 2 plots the difference-in-differences in the cumulative distribution function of years of education.
- The effect rises up to about six years of education, consistent with increased primary completion.
- There is some effect on lower secondary completion, but little evidence of strong effects at higher education levels.
- Figure 3 overlays education and log wage cohort-interaction coefficients.
- The wage pattern is noisier, but it rises for cohorts that also show education gains.

Detailed interpretation: Figure 2 answers an important question: did the program increase average schooling by moving children into primary school, or did it generate broader schooling gains at many levels? The answer is mainly primary. Figure 3 provides a visual link between the first stage and the reduced-form wage effect. The wage curve is less precise because wages are observed only for wage earners and because earnings are inherently noisier, but the pattern is broadly consistent with the education channel.

Economic message: The policy primarily increased basic schooling. The labor-market effects appear to operate through the same cohorts and levels of education targeted by the construction program.

TABLE 6—PROGRAM EFFECT AND RETURNS TO EDUCATION BY CATEGORIES OF REGION OF BIRTH

	Characteristics of region of birth						
	Density ^a		1976 Poverty ^b		Preprogram education ^c		
	Whole sample (1)	<Median (2)	>Median (3)	High (4)	Low (5)	<Median (6)	>Median (7)
<i>Panel A: Effect of the Program on Education</i>							
Dependent variable: Years of education. Sample: individuals ages 2 to 6 or 12 to 17 in 1974							
Interaction (2–6 in 1974)*program intensity in region of	0.15 (0.026)	0.19 (0.035)	-0.014 (0.048)	0.13 (0.058)	0.083 (0.035)	0.14 (0.040)	0.13 (0.036)
<i>Panel B: Effect of the Program on Wages</i>							
Dependent variable: log(hourly wage). Sample: individuals ages 2 to 6 or 12 to 17 in 1974 (wage earners)							
Interaction (2–6 in 1974)*program intensity in region of	0.017 (0.0074)	0.032 (0.011)	-0.00084 (0.012)	0.051 (0.017)	-0.00083 (0.0094)	0.028 (0.013)	0.0046 (0.0095)
<i>Panel C: Returns to Education</i>							
Dependent variable: log(hourly wage). Sample: wage earners							
Years of education	0.078 (0.00062)	0.11 (0.026)	No First stage (0.86]	0.10 (0.028)	No First stage (0.88]	0.12 (0.032)	0.029 (0.052)

Notes: Region of birth dummies, year of birth dummies, and interactions of year of birth dummies and the number of children and the enrollment in the region in 1971 are included in the regressions. Standard errors are in parentheses, F -statistics of the overidentification test are in square brackets.

^a The median density (the density for the region of birth for the median person in the weighted sample) is 308 inhabitants per square kilometer.

^b The high poverty provinces are the provinces where the proportion of people consuming less than 1,500 Rp per capita is larger than the national for rural regions (in the 1976 SUSENAS). I define “high poverty” as rural districts in these provinces which are: Lamuno, Central East Java, East Nusa Tenggara, Central Sulawesi, South Sulawesi, Southeast Sulawesi.

Table 6: Program Effect and Returns to Education by Categories of Region of Birth

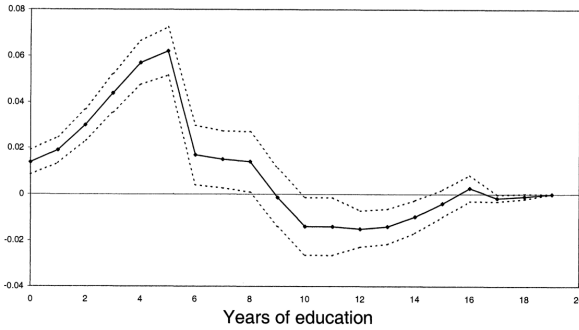


FIGURE 2. DIFFERENCE IN DIFFERENCES IN CDF (ESTIMATED FROM LINEAR PROBABILITY MODEL) WITH 95-PERCENT CONFIDENCE INTERVAL

(a) Figure 2: Difference in differences in CDF of education



(b) Figure 3: Age-in-1974 interactions in wage and education equations

7.7 Table 7: Effect of Education on Labor Market Outcomes: OLS and 2SLS Estimates

Read:

- Table 7 compares OLS and 2SLS estimates of returns to education.
- OLS log hourly wage returns are about 7.7 percent.
- 2SLS log hourly wage returns range from about 6.75 percent to 10.6 percent in the main specifications.
- The two-stage estimates using age-by-program instruments are broadly similar to OLS.
- The program also appears to increase participation in the wage sector.
- Completed-sample monthly earnings estimates are smaller, suggesting wage-sector returns may exceed returns for the self-employed.

Detailed interpretation: This is the main returns-to-education table. The similarity of OLS and IV estimates is notable. It implies that, in this setting, cross-sectional OLS estimates may not be sharply upward biased by omitted family and community background. However, the 2SLS estimate is local: it applies to those whose schooling was changed by primary school construction, not necessarily to returns at all education levels.

Economic message: The return to marginal schooling induced by school construction is economically meaningful and comparable to many estimates from richer countries.

7.8 Table 8: Evaluation of the Program's Net Return

Read:

- Table 8 converts the wage estimates into a social cost-benefit calculation.
- It considers deadweight-loss coefficients of 0.2 and 0.6.
- Depending on assumptions, annual benefits exceed costs by the mid-to-late 1990s.
- Discounted net benefits through 2050 can be large under moderate growth assumptions.
- Internal rates of return vary by assumptions but can be around 7.5 to 11.8 percent under plausible growth scenarios.

TABLE 7—EFFECT OF EDUCATION ON LABOR MARKET OUTCOMES: OLS AND 2SLS ESTIMATES

Method	Instrument	(1)	(2)	(3)	(4)
<i>Panel A: Sample of Wage Earners</i>					
<i>Panel A1: Dependent variable: log(hourly wage)</i>					
OLS		0.0776 (0.000620)	0.0777 (0.000621)	0.0767 (0.000646)	
2SLS	Year of birth dummies*program intensity in region of birth	0.0675 (0.0280) [0.96]	0.0809 (0.0272) [0.9]	0.106 (0.0222) [0.93]	0.0908 (0.0541) [0.9]
2SLS	(Aged 2–6 in 1974)*program intensity in region of birth	0.0752 (0.0338) (0.0338)	0.0862 (0.0336) (0.0336)	0.104 (0.0304) (0.0304)	
<i>Panel A2: Dependent variable: log(monthly earnings)</i>					
OLS		0.0698 (0.000601)	0.0698 (0.000602)	0.0689 (0.000628)	
2SLS	Year of birth dummies*program intensity in region of birth	0.0756 (0.0280) [0.73]	0.0925 (0.0278) [0.63]	0.0913 (0.0219) [0.58]	0.134 (0.0631) [0.7]
<i>Panel B: Whole Sample</i>					
<i>Panel B1: Dependent variable: participation in the wage sector</i>					
OLS		0.0328 (0.00311)	0.0327 (0.000311)	0.0337 (0.000319)	
2SLS	Year of birth dummies*program intensity in region of birth	0.101 (0.0210) [0.66]	0.118 (0.0197) [0.93]	0.0892 (0.0162) [1.12]	
<i>Panel B2: Dependent variable: log(monthly earnings), imputed for self-employed individuals</i>					
OLS		0.0539 (0.000354)	0.0539 (0.000354)	0.0539 (0.000355)	
2SLS	Year of birth dummies*program intensity in region of birth	0.0509 (0.0157) [0.68]	0.0745 (0.0136) [0.58]	0.0346 (0.0138) [1.16]	
<i>Control variables:</i>					
Year of birth*enrollment rate in 1971		No	Yes	Yes	Yes
Year of birth*water and sanitation program		No	No	Yes	No
Propensity score, propensity score squared		No	No	No	Yes

Notes: Year of birth dummies, region of birth dummies, and the interactions between year of birth dummies and the number of children in the region of birth in 1971 are included in the regressions. Standard errors are in parentheses. *F*-statistics of the test of overidentification restrictions are in square brackets.

TABLE 8—EVALUATION OF THE PROGRAM'S NET RETURN

	Deadweight loss coefficient			
	0.2		0.6	
	(1)	(2)	(3)	(4)
<i>Panel A: Results</i>				
Control for year of birth*enrollment rate	No	Yes	No	Yes
First year where benefit > costs (discount rate = 5 percent)				
In annual value	1996	1996	1997	1997
In discounted sum	2005	2002	2009	2005
Discounted sum of net benefits in 2050 (growth rate after 1997 = 5 percent, discount rate 5 percent)				
In million 1990 U.S.\$	13,025	13,096	11,340	18,807
As a fraction of Indonesia's GDP in 1973	0.30	0.36	0.31	0.52
Divided by initial costs	24.1	24.2	21.0	35.0
Discounted sum of net benefits in 2050 (growth rate after 1997 = 2 percent, discount rate 5 percent)				
In million 1990 U.S.\$	6,691	11,589	5,008	9,905
As a fraction of Indonesia's GDP in 1973	0.18	0.32	0.14	0.27
Divided by initial costs	12.4	21.4	9.26	18.3
Discounted sum of net benefits in 2050 (growth rate from 1973 = 2 percent, discount rate 5 percent)				
In million 1990 U.S.\$	−631.6	1,200	−2,315	−483
As a fraction of Indonesia's GDP in 1973	−0.017	0.033	−0.063	−0.013
Divided by initial costs	−1.16	2.22	−4.28	−0.89
<i>Internal rate of return*</i>				
Growth rate after 1997 = 5 percent	0.102	0.118	0.0895	0.105
Growth rate after 1997 = 2 percent	0.088	0.106	0.0750	0.0915
Growth rate from 1973 = 2 percent	0.0443	0.059	0.0326	0.0467
<i>Panel B: Assumptions and Parameters</i>				
Population growth rate after 1997	0.015			
Yearly teacher's salary in 1973 (1990 U.S. dollars)	363			
Yearly teacher's salary in 1995 (1990 U.S. dollars)	2,467			
Total recurrent costs/teacher salary	1.25			
Total cost of construction (million 1990 U.S. dollars)	522			
Number of schools constructed	61,800			
Lifetime of the schools (years)	20			
Share of labor income in GDP	0.7			

Notes: The estimates underlying these calculations are taken from Table 5 (columns (7) and (8)). Program effect has been set to 0 for children aged 7 or older in 1974.

*The internal rate of return is the interest rate such that the net present value of the project at infinity is 0.

Table 7: Effect of Education on Labor Market Outcomes: OLS and 2SLS Estimates

Table 8: Evaluation of the Program's Net Return

- Under pessimistic growth assumptions, the net present value can be small or negative.

Detailed interpretation: The table is not just an accounting exercise; it links micro estimates to public finance. The economic value of school construction depends heavily on how wage gains are projected, what discount rate is used, and the deadweight cost of taxation. The exercise is conservative because it uses wages as the main benefit and excludes many non-wage returns to education. At the same time, it depends on the assumption that wage gains reflect productivity gains and not redistribution.

Economic message: Even a costly nationwide school construction program can produce high social returns if it raises human capital and labor productivity. The conclusion is robust in many, but not all, parameter scenarios.

8 Result map

- **Program allocation.** INPRES schools were targeted to low-enrollment areas; the targeting is strong and observable.
- **First stage.** Young cohorts in high-intensity regions gained about 0.12 to 0.19 years of schooling per school per 1,000 children.
- **Reduced-form wages.** The same exposure raised wages by roughly 1.5 to 2.7 percent.
- **Placebo timing.** Older cohorts show little effect, supporting the causal interpretation.
- **Mechanism.** Effects are concentrated around primary completion and in low-density regions where access constraints were stronger.
- **Returns.** IV estimates of returns to education are around 6.8 to 10.6 percent.
- **Cost-benefit.** Under plausible assumptions, the program generated positive net social returns.

9 Short summary

- The paper studies the Indonesian Sekolah Dasar INPRES primary school construction program.
- More than 61,000 schools were built between 1973-1974 and 1978-1979.
- Exposure depends on district-level school construction and age when the program began.
- The paper uses region-of-birth and year-of-birth fixed effects plus program-intensity-by-cohort interactions.
- High-program regions were initially less educated, so the design must compare cohort changes rather than levels.
- The program raised schooling by roughly 0.12 to 0.19 years per school per 1,000 children.
- It raised wages by roughly 1.5 to 2.7 percent for exposed cohorts.
- The implied return to schooling is about 6.8 to 10.6 percent.
- Effects are concentrated at primary education levels and among cohorts young enough to benefit.
- The results are robust to controls for initial enrollment and other development programs.
- Sample-selection checks show somewhat smaller returns outside the wage sector but do not overturn the main findings.
- Cost-benefit calculations suggest positive social returns under many plausible assumptions.
- The paper is a landmark quasi-experimental study of school construction and returns to education in a developing country.

10 Conclusion

10.1 Core conclusion

Duflo shows that a major school construction program in Indonesia increased educational attainment and later wages for cohorts young enough to benefit. The evidence supports the view that school availability can be a binding constraint on human capital accumulation and that expanding primary school infrastructure can generate economically meaningful returns.

10.2 Economic intuition

The intuition is straightforward. When schools are physically scarce, children face higher travel costs, overcrowding, or lack of available seats. Building schools lowers these access costs. Children who are still young enough to attend primary school acquire more education. Later, this additional schooling raises labor-market productivity and wages.

10.3 Identification lesson

The paper's most important methodological lesson is that a national program can be evaluated if exposure differs by geography and by cohort. Region fixed effects handle permanent differences across districts. Cohort fixed effects handle national changes across birth years. The interaction of program intensity and age-based exposure isolates the policy effect.

10.4 Incremental methodological contribution

The paper advances development economics by showing how to use government program rules and demographic timing to construct credible instruments for education. It also demonstrates the value of visual cohort-pattern tests: the estimated effects appear for exactly the cohorts that should be affected and not for older cohorts.

10.5 Policy implications

The results support public investment in basic education infrastructure, especially in areas where physical access is limited. They also suggest that the returns to such investments can be high enough to justify large public outlays. However, the program worked in a context where primary school demand was strong and broader government support for education existed, so the policy implication is not simply that school construction always works everywhere.

10.6 Limitations

The paper cannot fully measure national general equilibrium effects, such as a reduction in returns to education from increasing the supply of educated workers. It estimates local returns for individuals induced to get more primary schooling, not necessarily returns to secondary or higher education. The wage data also raise selection issues because wages are only observed for wage earners, though the paper addresses this carefully.

10.7 Takeaway

The paper's lasting contribution is to turn a large historical school-building program into a credible causal research design. It shows that infrastructure policy can have long-run human-capital and earnings effects, and it provides a benchmark for evaluating education investment in developing economies.